

# boba\_extension\_dim08

## Dimension 08: Web Page DOM Scraping & Dynamic Content Detection

### Research Objective

Comprehensive strategies for finding torrent and magnet links on any web page, including handling dynamically loaded content in SPAs, iframes, infinite scroll, and Shadow DOM scenarios.

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# 1. Content Script Injection Timing

## 1.1 Three Timing Options

Content scripts support three `run_at` values that determine when JavaScript is injected:

| Value                       | Description                                        | Use Case                                       |
|-----------------------------|----------------------------------------------------|------------------------------------------------|
| <code>document_start</code> | DOM is still loading, before any other scripts run | Early setup, must wait for DOM manually        |
| <code>document_end</code>   | DOM is complete, but subresources still loading    | Balance between early access and DOM readiness |
| <code>document_idle</code>  | DOM and resources fully loaded (default)           | Most content scripts, guaranteed DOM ready     |

**Claim:** The default `document_idle` is the recommended timing for most extensions. **Source:** Chrome for Developers - Content Scripts **URL:** <https://developer.chrome.com/docs/extensions/develop/concepts/content-scripts> **Date:** 2012-09-17 **Excerpt:** "document\_idle" ... The browser chooses a time to inject scripts between "document\_end" and immediately after the window.onload event fires... Use "document\_idle" whenever possible. **Confidence:** high

## 1.2 Handling Different readyState

```
/**
 * Robust initialization that works regardless of when content
 * script loads.
 * Handles all three run_at scenarios: document_start,
 * document_end, document_idle
 */
function initializeWhenReady(callback) {
  if (document.readyState === 'loading') {
    // DOM still loading - wait for complete
    document.addEventListener('DOMContentLoaded', callback);
  } else if (document.readyState === 'interactive' ||
    document.readyState === 'complete') {
    // DOM ready - can run immediately
    callback();
  }
}
```

```
// Usage
initializeWhenReady(() => {
  performInitialScan();
  setupMutationObserver();
});
```

## 1.3 Manifest Configuration

```
{
  "content_scripts": [
    {
      "matches": ["<all_urls>"],
      "js": ["content.js"],
      "run_at": "document_idle",
      "all_frames": false
    }
  ]
}
```

## 1.4 Handling Dynamic SPAs (like Gmail)

**Claim:** For heavily dynamic SPAs like Gmail, `document_idle` may not be sufficient because the DOM continues to change after `window.onload`.

**Source:** GMass Blog - Gmail Extension Timing **URL:** <https://www.gmass.co/blog/timing-gmail-chrome-extension-content-script/>

**Date:** 2020-12-12 **Excerpt:**

The confusion for me... is that the rules for Gmail are different... As long as you're wrapped in a `window.onload` if/then checking system, you're good. **Confidence:** high

```
/**
 * SPA-safe initialization that handles both immediate and
 * deferred DOM readiness.
 * Essential for Gmail, Google Docs, and other heavy SPAs.
 */
function safeInitialize(callback) {
  if (document.readyState === 'complete') {
    callback();
  } else {
    window.addEventListener('load', callback);
  }
}
```

---

## 2. DOM Traversal Strategies

### 2.1 Basic querySelectorAll Approach

The most efficient method for finding link elements with torrent/magnet content:

```
/**
 * Fast initial scan using CSS selectors - O(1) for selector
 * matching.
 * This is the most performant approach for finding <a> tags.
 */
function findLinksBySelectors() {
  const results = [];

  // Magnet links in href attributes
  const magnetLinks =
    document.querySelectorAll('a[href^="magnet:"]');
  magnetLinks.forEach(el => results.push({ element: el, type:
    'magnet', source: 'href' }));

  // .torrent file links
  const torrentLinks =
    document.querySelectorAll('a[href$=".torrent"]');
  torrentLinks.forEach(el => results.push({ element: el, type:
    'torrent', source: 'href' }));

  // Data attributes that may contain magnet links
  const dataMagnetLinks = document.querySelectorAll('[data-
    magnet]');
  dataMagnetLinks.forEach(el => results.push({
    element: el,
    type: 'magnet',
    source: 'data-attribute',
    value: el.getAttribute('data-magnet')
  }));

  return results;
}
```

### 2.2 TreeWalker for Text Node Traversal

**Claim:** TreeWalker is a DOM API that provides efficient traversal of specific node types and can be faster than recursive manual DOM walking for certain operations. **Source:** MDN - TreeWalker **URL:** <https://developer.mozilla.org/en-US/docs/Web/API/TreeWalker> **Date:** N/A **Excerpt:** A TreeWalker object represents the nodes of a document subtree and a position within them. **Confidence:** high

```

/**
 * Uses TreeWalker to efficiently scan for text nodes containing
 * magnet links.
 * TreeWalker is optimized by the browser and handles edge cases
 * automatically.
 *
 * @param {Node} rootNode - Root node to scan (default:
 *   document.body)
 * @returns {Array<{textNode, parentElement, magnetUrls}>}
 */
function findMagnetLinksInTextNodes(rootNode = document.body) {
  const results = [];
  const magnetRegex = /magnet:\?xt=urn:btih:[a-f0-9]{32,40}
    [^"'<>]*/gi;

  const walker = document.createTreeWalker(
    rootNode,
    NodeFilter.SHOW_TEXT,
    {
      acceptNode: (node) => {
        // Skip text nodes inside script/style tags
        const parent = node.parentElement;
        if (!parent) return NodeFilter.FILTER_SKIP;
        const parentTag = parent.tagName.toLowerCase();
        if (parentTag === 'script' || parentTag === 'style' ||
          parentTag === 'noscript') {
          return NodeFilter.FILTER_SKIP;
        }
        // Skip empty or whitespace-only nodes
        if (!node.nodeValue || !node.nodeValue.trim()) {
          return NodeFilter.FILTER_SKIP;
        }
        return NodeFilter.FILTER_ACCEPT;
      }
    },
    false
  );

  let textNode;
  while ((textNode = walker.nextNode()) !== null) {
    const matches = textNode.nodeValue.match(magnetRegex);
    if (matches && matches.length > 0) {
      results.push({
        textNode,
        parentElement: textNode.parentElement,
        magnetUrls: matches
      });
    }
  }

  return results;
}

```

## 2.3 Recursive DOM Traversal (Shadow DOM Compatible)

```
/**
 * Recursive DOM traversal that handles Shadow DOM boundaries.
 * More flexible than TreeWalker but requires manual
 *   implementation.
 *
 * @param {Node} root - Starting node
 * @param {Function} callback - Called for each element
 * @param {Object} options - { includeShadowDOM: boolean,
 *   maxDepth: number }
 */
function traverseDOM(root, callback, options = {}) {
  const { includeShadowDOM = true, maxDepth = 100 } = options;
  const visited = new Set();

  function walk(node, depth) {
    if (depth > maxDepth) return;
    if (!node) return;
    if (visited.has(node)) return; // Prevent infinite loops
    visited.add(node);

    if (node.nodeType === Node.ELEMENT_NODE) {
      callback(node);

      // Traverse children in light DOM
      if (node.children) {
        for (const child of node.children) {
          walk(child, depth + 1);
        }
      }

      // Traverse shadow DOM if present
      if (includeShadowDOM && node.shadowRoot) {
        walk(node.shadowRoot, depth + 1);
      }
    }

    // Also traverse childNodes for non-element nodes (text,
    //   comments)
    if (node.childNodes) {
      for (const child of node.childNodes) {
        if (child.nodeType !== Node.ELEMENT_NODE) {
          walk(child, depth + 1);
        }
      }
    }
  }

  walk(root, 0);
}
```

---

## 3. Link Element Detection Patterns

### 3.1 Comprehensive Link Scanner

```
/**
 * Complete link scanner that checks all common patterns for
 * torrent/magnet links.
 * Designed to be extensible with site-specific selectors.
 */
class LinkScanner {
  constructor() {
    // Magnet URI detection regex - matches the xt=urn:btih:
    // prefix with hash
    this.magnetRegex = /magnet:\?xt=urn:btih:[a-f0-9]{32,40}
    [^'\s<>]*/gi;

    // Site-specific CSS selectors for known torrent sites
    this.siteSelectors = {
      'thepiratebay': {
        magnet: ['a[href^="magnet:"]', '.download
        a[href^="magnet:"]'],
        torrent: ['a[href$=".torrent"]'],
        details: { name: '#title', seeders: 'dt:first-child +
        dd' }
      },
      '1337x': {
        magnet: ['a[href^="magnet:"]', '.torrent-detail-page
        a[href^="magnet:"]'],
        torrent: ['a[href$=".torrent"]'],
        details: { name: '.box-info-heading h1' }
      },
      'nyaa': {
        magnet: ['a[href^="magnet:"]', '.torrent-download-
        link[href^="magnet:"]'],
        torrent: ['a[href$=".torrent"]']
      },
      'rutracker': {
        magnet: ['a[href^="magnet:"]', '.magnet-
        link[href^="magnet:"]'],
        torrent: ['a[href$=".torrent"]']
      }
    };
  }

  /**
   * Scan all <a> tags for magnet/torrent hrefs
   */
  scanAnchorLinks(container = document) {
    const results = [];
    const links = container.querySelectorAll('a');

    for (const link of links) {
```

```

    const href = link.getAttribute('href');
    if (!href) continue;

    // Check for magnet protocol
    if (href.startsWith('magnet:')) {
        results.push({
            element: link,
            type: 'magnet',
            url: href,
            text: link.textContent.trim(),
            title: link.getAttribute('title') || ''
        });
        continue;
    }

    // Check for .torrent file
    if (href.endsWith('.torrent') ||
        href.includes('.torrent?')) {
        results.push({
            element: link,
            type: 'torrent-file',
            url: href,
            text: link.textContent.trim()
        });
        continue;
    }

    // Check for data attributes
    const dataMagnet = link.getAttribute('data-magnet');
    if (dataMagnet) {
        results.push({
            element: link,
            type: 'magnet',
            url: dataMagnet,
            source: 'data-attribute'
        });
    }
}

return results;
}

/**
 * Check for onclick handlers that open magnet links
 */
scanClickHandlers(container = document) {
    const results = [];
    const elements = container.querySelectorAll('[onclick]');

    for (const el of elements) {
        const onclick = el.getAttribute('onclick') || '';
    }
}

```



```

const magnetMatch = onclick.match(/magnet:\?xt=urn:btih:[a-
f0-9]{32,40}[^"\s]*/i);
if (magnetMatch) {
  results.push({
    element: el,
    type: 'magnet',
    url: magnetMatch[0],
    source: 'onclick-handler'
  });
}
}
return results;
}
}

```

---

## 4. Text-Based Magnet Detection

### 4.1 Regex Pattern for Magnet Links

**Claim:** A magnet link regex must match the magnet:?xt=urn:btih: prefix followed by a 32-40 character hexadecimal hash, with optional parameters (dn, tr, xl, etc.) **Source:** magnet-link-regex npm package **URL:** <https://github.com/tiaanduplessis/magnet-link-regex/blob/master/index.js> **Date:** 2018-06-04 **Excerpt:** /magnet:\?xt=urn:[a-z0-9]+:[a-z0-9]{32,40}&dn=.\+&tr=.\+/gi **Confidence:** high

```

/**
 * Comprehensive regex patterns for magnet link detection.
 * Uses multiple patterns to handle different magnet link formats.
 */
const MAGNET_PATTERNS = {
  // Full magnet link with BTIH hash (hex, 32-40 chars)
  // Matches: magnet:?xt=urn:btih:HASH&dn=NAME&tr=TRACKER...
  FULL: /magnet:\?xt=urn:btih:[a-fA-F0-9]{32,40}(?:&[\s"'<>]+)*/gi,

  // Minimal magnet link (just hash)
  // Matches: magnet:?xt=urn:btih:HASH
  MINIMAL: /magnet:\?xt=urn:btih:[a-fA-F0-9]{32,40}/gi,

  // Magnet link with any URN type (btih, sha1, ed2k, etc.)
  // Matches: magnet:?xt=urn:TYPE:HASH
  URN_GENERIC: /magnet:\?xt=urn:[a-z0-9]+:[a-zA-Z0-9]{32,40}(?:&[\s"'<>]+)*/gi,

  // Extract BTIH hash specifically
  // Captures group 1 as the hash

```

```

BTIH_EXTRACT: /xt=urn:btih:([a-fA-F0-9]{32,40})/i,

// Extract display name (dn parameter)
DN_EXTRACT: /[?&]dn=([^&]+)/i,

// Extract tracker URLs (tr parameters)
TR_EXTRACT: /[?&]tr=([^&]+)/gi
};

/**
 * Validates a magnet link string and extracts components.
 *
 * @param {string} url - The magnet URL to validate
 * @returns {Object|null} Parsed magnet info or null if invalid
 */
function parseMagnetLink(url) {
  if (!url || !url.startsWith('magnet:')) return null;

  const btihMatch = url.match(MAGNET_PATTERNS.BTIH_EXTRACT);
  if (!btihMatch) return null;

  const hash = btihMatch[1].toLowerCase();

  // Validate hash length (32 = base32, 40 = hex)
  if (hash.length !== 32 && hash.length !== 40) return null;

  const dnMatch = url.match(MAGNET_PATTERNS.DN_EXTRACT);
  const trackers = [];
  let trMatch;
  while ((trMatch = MAGNET_PATTERNS.TR_EXTRACT.exec(url)) !==
    null) {
    try { trackers.push(decodeURIComponent(trMatch[1])); }
    catch(e) { trackers.push(trMatch[1]); }
  }
  MAGNET_PATTERNS.TR_EXTRACT.lastIndex = 0; // Reset regex

  return {
    hash,
    hashType: hash.length === 40 ? 'hex' : 'base32',
    name: dnMatch ? decodeURIComponent(dnMatch[1]) : '',
    trackers,
    raw: url
  };
}

```

## 4.2 Text Node Scanner

```

/**
 * Scans all text nodes on the page for magnet links that aren't
 *   in <a> tags.
 * This finds magnet links displayed as plain text, in <code>,
 *   <pre>, <span>, etc.

```

```

*
* @param {HTMLElement} container - Container to scan
* @returns {Array<{element: HTMLElement, text: string, magnets:
    Array<string>}>}
*/
function scanTextNodesForMagnets(container = document.body) {
    const results = [];
    const processedParents = new Set(); // Avoid duplicates

    const walker = document.createTreeWalker(
        container,
        NodeFilter.SHOW_TEXT,
        null,
        false
    );

    let textNode;
    while ((textNode = walker.nextNode()) !== null) {
        const parent = textNode.parentElement;
        if (!parent) continue;

        // Skip if parent is a link (already handled by
        // scanAnchorLinks)
        if (parent.tagName === 'A' &&
            parent.getAttribute('href')?.startsWith('magnet:'))
            continue;

        // Skip script/style/noscript content
        const skipTags = ['SCRIPT', 'STYLE', 'NOSCRIPT', 'TEMPLATE'];
        if (skipTags.includes(parent.tagName)) continue;

        const text = textNode.nodeValue;
        const matches = text.match(MAGNET_PATTERNS.FULL);

        if (matches && matches.length > 0) {
            if (!processedParents.has(parent)) {
                processedParents.add(parent);
                results.push({
                    element: parent,
                    text: text.trim(),
                    magnets: matches
                });
            }
        }
    }

    return results;
}

```

---

## 5. Magnet Link URI Patterns

### 5.1 Magnet URI Specification

**Claim:** The Magnet URI scheme defines specific parameters: xt (exact topic), dn (display name), tr (tracker), xl (exact length), kt (keyword topic), ws (web seed), xs (exact source), as (acceptable source), mt (manifest topic). **Source:** Baiduwiki - Magnet URI scheme **URL:** <https://baike.baidu.com/en/item/Magnet%20URI%20scheme/1415756> **Date:** N/A **Excerpt:** xt=urn:btih:[BitTorrent Info Hash (Hexadecimal)]...

This is the identifier of the file and is indispensable.

**Confidence:** high

### 5.2 Magnet URI Components Reference

| Parameter | Name              | Required | Description                              |
|-----------|-------------------|----------|------------------------------------------|
| xt        | Exact Topic       | Yes      | URN with file hash (e.g., urn:btih:HASH) |
| dn        | Display Name      | No       | Filename shown to user                   |
| tr        | Tracker           | No       | Tracker URL for peer discovery           |
| xl        | Exact Length      | No       | File size in bytes                       |
| kt        | Keyword Topic     | No       | Search keywords                          |
| ws        | Web Seed          | No       | HTTP/FTP seed URL                        |
| xs        | Exact Source      | No       | P2P source address                       |
| as        | Acceptable Source | No       | Fallback web server URL                  |
| mt        | Manifest Topic    | No       | Link to metafile with magnet list        |
| so        | Select Only       | No       | Specific file indices to download        |

## 5.3 Modern Magnet Link Formats

**Claim:** BitTorrent v3.1 introduces new hash algorithms beyond SHA-1, with magnet links using format `magnet:?xt=urn:btih-SPECIFIER:hash`.

**Source:** Tixati Protocol v3.1 Specification **URL:** <https://tixati.com/specs/bittorrent/v3.1> **Date:** 2025-10-07 **Excerpt:** Example: `magnet:?xt=urn:btih-`

`sha3:hkgkj2kv6g5kjmps66ec3tqbeekuojlrlnlig3bifqmsuyxmw64ua&dn=somef`

`ilename.jpg` **Confidence:** medium

---

## 6. .torrent File Link Patterns

### 6.1 Detection Patterns

```
/**
 * Comprehensive .torrent link detection.
 */
const TORRENT_PATTERNS = {
  // Direct .torrent file link
  DIRECT: /\\.torrent(?:\?.*)?$/i,

  // Download endpoints that serve .torrent files
  DOWNLOAD_ENDPOINTS: [
    /\download\.php\?.*torrent/i,
    /\torrent\download\//i,
    /\download\torrent\//i,
    /\torrents\download\//i,
    /\get\torrent\//i
  ],

  // Common file hosting patterns for torrents
  FILE_HOSTS: [
    /torcache\.net/i,
    /itorrents\.org/i,
    /zoink\.it/i,
    /googleapis\.com\.*\torrent/i,
    /rarbg\.to\.*\torrent/i
  ]
};

/**
 * Check if a URL points to a .torrent file or torrent download
 * endpoint.
 */
function isTorrentUrl(url) {
  if (!url) return false;
```

```

// Direct .torrent file
if (TORRENT_PATTERNS.DIRECT.test(url)) return true;

// Download endpoints
for (const pattern of TORRENT_PATTERNS.DOWNLOAD_ENDPOINTS) {
  if (pattern.test(url)) return true;
}

// Known torrent file hosts
for (const pattern of TORRENT_PATTERNS.FILE_HOSTS) {
  if (pattern.test(url)) return true;
}

return false;
}

```

---

## 7. MutationObserver for Dynamic Content

### 7.1 Basic MutationObserver Setup

**Claim:** MutationObserver fires its callback on the microtask queue, making it ~88x faster than polling with `setTimeout` for DOM change detection. **Source:** Use a MutationObserver to Handle DOM Nodes That Don't Exist Yet **URL:** <https://macarthur.me/posts/use-mutation-observer-to-handle-nodes-that-dont-exist-yet> **Date:** 2023-03-13 **Excerpt:** polling with a zero-delay `setTimeout()` ... was around 88 times slower than the MutationObserver **Confidence:** high

```

/**
 * Sets up a MutationObserver to detect dynamically added content.
 * Uses debouncing to batch rapid DOM changes and avoid excessive
 * scanning.
 *
 * @param {Function} onChange - Callback when DOM changes are
 *   detected
 * @param {Object} options - Configuration options
 * @returns {MutationObserver} The configured observer
 */
function setupDynamicContentObserver(onChange, options = {}) {
  const {
    debounceMs = 500, // Wait for DOM changes to settle
    maxBatchSize = 100, // Max mutations to process at once
    subtree = true, // Watch entire DOM tree
    attributes = false, // Watch attribute changes
                        (performance cost)
  } = options;

  let debounceTimer = null;

```

```

let pendingMutations = [];
let isProcessing = false;

const observer = new MutationObserver((mutations) => {
  // Filter relevant mutations (added nodes only)
  const relevantMutations = mutations.filter(m =>
    m.type === 'childList' && m.addedNodes.length > 0
  );

  if (relevantMutations.length === 0) return;

  pendingMutations.push(...relevantMutations);

  // Limit pending mutations to prevent memory issues
  if (pendingMutations.length > maxBatchSize) {
    pendingMutations = pendingMutations.slice(-maxBatchSize);
  }

  // Debounce the scan to batch rapid changes
  clearTimeout(debounceTimer);
  debounceTimer = setTimeout(() => {
    if (isProcessing) return;
    isProcessing = true;

    try {
      // Collect all added nodes
      const addedNodes = [];
      for (const mutation of pendingMutations) {
        for (const node of mutation.addedNodes) {
          if (node.nodeType === Node.ELEMENT_NODE) {
            addedNodes.push(node);
          }
        }
      }
      pendingMutations = [];

      if (addedNodes.length > 0) {
        onChange(addedNodes);
      }
    } finally {
      isProcessing = false;
    }
  }, debounceMs);
});

// Observe the document body for changes
observer.observe(document.body, {
  childList: true,
  subtree: subtree,
  attributes: attributes
});

```

```

    return observer;
}

```

## 7.2 Incremental Scanning

```

/**
 * Incremental scanner that only scans newly added nodes.
 * Much more efficient than re-scanning the entire DOM.
 *
 * @param {Array<Node>} addedNodes - Nodes added by
 *   MutationObserver
 * @returns {Array<Object>} Found torrent/magnet links
 */
function incrementalScan(addedNodes) {
    const scanner = new LinkScanner();
    const results = [];

    for (const node of addedNodes) {
        // The node itself might be a matching element
        if (node.nodeType === Node.ELEMENT_NODE) {
            // Scan the element itself
            results.push(...scanner.scanAnchorLinks(node));
            results.push(...scanner.scanClickHandlers(node));

            // Scan text nodes within the element
            const textResults = scanTextNodesForMagnets(node);
            for (const tr of textResults) {
                for (const magnet of tr.magnets) {
                    results.push({
                        element: tr.element,
                        type: 'magnet',
                        url: magnet,
                        source: 'text-node'
                    });
                }
            }

            // Check for shadow DOM
            if (node.shadowRoot) {
                results.push(...scanner.scanAnchorLinks(node.shadowRoot));
                results.push(...scanTextNodesForMagnets(node.shadowRoot));
            }
        }
    }

    return results;
}

```



## 7.3 Infinite Scroll Detection

```
/**
 * Uses IntersectionObserver to detect when new content is being
 * loaded
 * (infinite scroll pattern). Complements MutationObserver for
 * scroll-based loading.
 *
 * @param {Function} onNearBottom - Called when user approaches
 * page bottom
 * @param {Object} options - Configuration
 */
function setupInfiniteScrollDetector(onNearBottom, options = {}) {
  const { rootMargin = '200px', threshold = 0 } = options;

  // Create a sentinel element at the bottom of the page
  let sentinel =
    document.getElementById('__torrent_scanner_sentinel');
  if (!sentinel) {
    sentinel = document.createElement('div');
    sentinel.id = '__torrent_scanner_sentinel';
    sentinel.style.cssText =
      'position:absolute;bottom:0;left:0;width:1px;height:1px;';
    document.body.appendChild(sentinel);
  }

  const observer = new IntersectionObserver((entries) => {
    if (entries[0].isIntersecting) {
      onNearBottom();
    }
  }, { rootMargin, threshold });

  observer.observe(sentinel);
  return observer;
}
```

---

## 8. iframe Handling

### 8.1 Cross-Origin iframe Limitations

**Claim:** Content scripts with `all_frames: true` are injected into all matching frames, but cross-origin iframes require special handling - content scripts can access their own DOM but cannot access the DOM of cross-origin parent/child frames from JavaScript. **Source:** MDN - content\_scripts manifest **URL:** [https://developer.mozilla.org/en-US/docs/Mozilla/Add-ons/WebExtensions/manifest.json/content\\_scripts](https://developer.mozilla.org/en-US/docs/Mozilla/Add-ons/WebExtensions/manifest.json/content_scripts) **Date:**

2025-11-26 **Excerpt:** "all\_frames": true ... This does not inject into child frames where only their parent matches the URL requirements and the child frame does not match. **Confidence:** high

## 8.2 iframe Detection Strategies

```
/**
 * Strategies for handling iframes that may contain torrent links.
 * Uses multiple approaches due to cross-origin restrictions.
 */
class IframeHandler {
  constructor() {
    this.knownIframes = new Map(); // iframe -> lastSrc
  }

  /**
   * Method 1: Check if content script is running inside an
   *           iframe.
   * Use this to adapt behavior when inside a frame context.
   */
  isInsideIframe() {
    return window.self !== window.top;
  }

  /**
   * Method 2: Try to access same-origin iframe content.
   * Only works if iframe is same-origin (limited practical use
   * for torrent sites).
   */
  tryAccessSameOriginIframes() {
    const results = [];
    const iframes = document.querySelectorAll('iframe');

    for (const iframe of iframes) {
      try {
        // This will throw if cross-origin
        const iframeDoc = iframe.contentDocument ||
          iframe.contentWindow?.document;
        if (iframeDoc) {
          // Same-origin iframe - can scan directly
          const scanner = new LinkScanner();
          results.push(...scanner.scanAnchorLinks(iframeDoc));

          results.push(...scanner.scanTextNodesForMagnets(iframeDoc.body));
        }
      } catch (e) {
        // Cross-origin iframe - cannot access
        // Content script with all_frames:true should handle this
      }
    }
  }
}
```

```

    return results;
}

/**
 * Method 3: Monitor iframe src changes using MutationObserver.
 * Detects when iframes are added or their source changes.
 */
monitorIframes(onIframeDetected) {
    const observer = new MutationObserver((mutations) => {
        for (const mutation of mutations) {
            for (const node of mutation.addedNodes) {
                if (node.nodeType === Node.ELEMENT_NODE) {
                    // Check if the node itself is an iframe
                    if (node.tagName === 'IFRAME') {
                        onIframeDetected(node);
                    }
                    // Check if the node contains iframes
                    const nestedIframes = node.querySelectorAll?.(
                        'iframe');
                    if (nestedIframes) {
                        for (const iframe of nestedIframes) {
                            onIframeDetected(iframe);
                        }
                    }
                }
            }
        }

        // Check for src attribute changes on existing iframes
        if (mutation.type === 'attributes' &&
            mutation.target.tagName === 'IFRAME' &&
            mutation.attributeName === 'src') {
            onIframeDetected(mutation.target);
        }
    });

    observer.observe(document.body, {
        childList: true,
        subtree: true,
        attributes: true,
        attributeFilter: ['src']
    });

    return observer;
}
}

```

### 8.3 Manifest Configuration for iframe Support

```

{
  "content_scripts": [

```

```

{
  "matches": ["<all_urls>"],
  "js": ["content.js"],
  "run_at": "document_idle",
  "all_frames": true,
  "match_about_blank": true
}
]
}

```

**Claim:** match\_about\_blank: true allows content scripts to run in about:blank iframes, which is useful for detecting dynamically created empty iframes that get populated with content. **Source:** MDN - content\_scripts manifest **URL:** [https://developer.mozilla.org/en-US/docs/Mozilla/Add-ons/WebExtensions/manifest.json/content\\_scripts](https://developer.mozilla.org/en-US/docs/Mozilla/Add-ons/WebExtensions/manifest.json/content_scripts) **Date:** 2025-11-26 **Excerpt:** "match\_about\_blank": true ... This is especially useful to run scripts in empty iframes, whose URL is "about:blank". **Confidence:** high

---

## 9. Shadow DOM Traversal

### 9.1 Understanding Shadow DOM Boundaries

**Claim:** document.querySelectorAll() does NOT return elements inside Shadow DOM. Elements in shadow trees are encapsulated and require explicit shadow root traversal. **Source:** MDN - Using shadow DOM **URL:** [https://developer.mozilla.org/en-US/docs/Web/API/Web\\_components/Using\\_shadow\\_DOM](https://developer.mozilla.org/en-US/docs/Web/API/Web_components/Using_shadow_DOM) **Date:** 2025-12-17 **Excerpt:** document.querySelector("p") // null ... document.querySelectorAll('\*') grabs all the elements in the DOM. With the shadow DOM, though, it doesn't work that way **Confidence:** high

### 9.2 Shadow DOM Piercing Implementation

```

/**
 * Recursively scans through Shadow DOM boundaries to find torrent
 * links.
 * Handles both open and (limited) closed shadow roots.
 *
 * @param {Node} root - Root node to scan from
 * @returns {Array<Object>} Found torrent/magnet links
 */
function scanThroughShadowDOM(root = document) {
  const results = [];

```

```

const scanner = new LinkScanner();
const visitedShadowRoots = new WeakSet();

function scanNode(node) {
  if (!node) return;

  if (node.nodeType === Node.ELEMENT_NODE) {
    // Scan this element's light DOM
    results.push(...scanner.scanAnchorLinks(node));
    results.push(...scanner.scanClickHandlers(node));

    // Check for shadow root
    const shadowRoot = node.shadowRoot;
    if (shadowRoot && !visitedShadowRoots.has(shadowRoot)) {
      visitedShadowRoots.add(shadowRoot);

      // Recursively scan the shadow DOM
      scanNode(shadowRoot);
    }

    // Recurse into children
    if (node.children) {
      for (const child of node.children) {
        scanNode(child);
      }
    }
  }

  // Handle DocumentFragment roots (shadow roots)
  if (node.nodeType === Node.DOCUMENT_FRAGMENT_NODE) {
    const children = node.children || node.querySelectorAll?.(
      ':scope > *');
    if (children) {
      for (const child of children) {
        scanNode(child);
      }
    }
  }
}

scanNode(root);
return results;
}

```

## 9.3 Using TreeWalker with Shadow DOM

```

/**
 * Alternative approach using NodeIterator to find elements with
 * shadow roots,
 * then scanning each shadow root separately.
 */

```

```

function findAllShadowHosts(root = document.body) {
  const shadowHosts = [];

  const nodeIterator = document.createNodeIterator(
    root,
    NodeFilter.SHOW_ELEMENT,
    (node) => node.shadowRoot ? NodeFilter.FILTER_ACCEPT :
      NodeFilter.FILTER_REJECT
  );

  let element;
  while ((element = nodeIterator.nextNode()) !== null) {
    shadowHosts.push(element);
  }

  return shadowHosts;
}

/**
 * Scans all shadow roots found in the document.
 */
function scanAllShadowRoots() {
  const results = [];
  const scanner = new LinkScanner();

  const shadowHosts = findAllShadowHosts();
  for (const host of shadowHosts) {
    if (host.shadowRoot) {
      results.push(...scanner.scanAnchorLinks(host.shadowRoot));
      results.push(...scanner.scanTextNodesForMagnets(host.shadowRoot));

      // Recursively check for nested shadow roots
      results.push(...scanAllShadowRoots.call(null,
        host.shadowRoot));
    }
  }

  return results;
}

```

## 9.4 Third-Party Library: kagekiri

**Claim:** kagekiri is a Salesforce library that implements `querySelectorAll` to traverse shadow DOM boundaries. **Source:** GitHub - salesforce/kagekiri **URL:** <https://github.com/salesforce/kagekiri> **Date:** 2019-08-07 **Excerpt:** Shadow DOM-piercing query APIs... kagekiri parses the CSS selector using `postcss-selector-parser`. Then it queries the entire DOM tree, traverses any `shadowRoots` it may find **Confidence:** high

```
// Using kagekiri for shadow DOM piercing (requires npm install
    kagekiri)
// import { querySelectorAll } from 'kagekiri';

// kagekiri.querySelectorAll('.container a[href^="magnet"]')
// would find magnet links even inside shadow DOM
```

---

## 10. Site-Specific Detection Patterns

### 10.1 Pattern Database

Based on analysis of popular torrent sites and userscript implementations:

```
/**
 * Site-specific selectors for known torrent sites.
 * These override generic selectors for better accuracy.
 */
const SITE_PATTERNS = {
  // The Pirate Bay patterns
  'thepiratebay': {
    domains: ['thepiratebay.org', 'thepiratebay.se',
      'piratebay.live', 'baymirror.com'],
    magnet: [
      'a[href^="magnet:"]', // Standard
      magnet links
      '.download a[href^="magnet:"]', // Download
      section
      'a[title^="Download this torrent using magnet"]' // TPB-
      specific title
    ],
    torrent: [
      'a[href$=".torrent"]',
      'a[href*="/torrent/"]'
    ],
    detailsPage: {
      magnet: '.download a[href^="magnet:"]',
      name: '#title',
      seeders: 'dt:contains("Seeders") + dd'
    }
  },
  // 1337x patterns
  '1337x': {
    domains: ['1337x.to', '1337x.st', 'x1337x.ws'],
    magnet: [
      'a[href^="magnet:"]',
      '.torrent-detail-page a[href^="magnet:"]',
      'a[href*="magnet:?xt=urn:btih:"]'
    ]
  }
}
```

```

    ],
    torrent: [
        'a[href$=".torrent"]',
        '.torrent-down' // Download button
    ]
},

// Nyaa (anime torrents)
'nyaa': {
    domains: ['nyaa.si', 'nyaa.net'],
    magnet: [
        'a[href^="magnet:"]',
        '.torrent-download-link[href^="magnet:"]',
        'a[href*="magnet:?xt=urn:btih:"]'
    ],
    torrent: [
        'a[href$=".torrent"]'
    ]
},

// RuTracker (Russian)
'rutracker': {
    domains: ['rutracker.org', 'rutracker.net', 'rutracker.nl'],
    magnet: [
        'a[href^="magnet:"]',
        '.magnet-link[href^="magnet:"]',
        '.dl-link[href^="magnet:"]'
    ],
    torrent: [
        'a[href$=".torrent"]',
        'a.dl-stub' // Download stub
    ]
},

// General torrent indexers
'generic': {
    magnet: [
        'a[href^="magnet:"]',
        'a[href*="magnet:?xt=urn:btih:"]'
    ],
    torrent: [
        'a[href$=".torrent"]',
        'a[href*=".torrent?"]',
        'a[href*="/download/"][href*=".torrent"]'
    ]
}
};

/**
 * Detect which site pattern to use based on current domain.
 */
function detectSitePattern() {

```



```

const hostname = window.location.hostname.toLowerCase();

for (const [siteName, pattern] of
  Object.entries(SITE_PATTERNS)) {
  if (pattern.domains) {
    for (const domain of pattern.domains) {
      if (hostname === domain || hostname.endsWith('.' +
        domain)) {
        return { siteName, pattern };
      }
    }
  }
}

// Default to generic pattern
return { siteName: 'generic', pattern: SITE_PATTERNS.generic };
}

```

## 10.2 Using Site-Specific Selectors

```

/**
 * Optimized scanner that uses site-specific selectors when
 * available.
 */
function scanWithSitePatterns(container = document) {
  const { siteName, pattern } = detectSitePattern();
  const results = [];

  // Use site-specific magnet selectors if available
  const magnetSelectors = pattern.magnet ||
    SITE_PATTERNS.generic.magnet;
  for (const selector of magnetSelectors) {
    try {
      const elements = container.querySelectorAll(selector);
      for (const el of elements) {
        const href = el.getAttribute('href');
        if (href && href.startsWith('magnet:')) {
          results.push({ element: el, type: 'magnet', url: href,
            site: siteName });
        }
      }
    } catch (e) {
      // Invalid selector, skip
    }
  }

  // Use site-specific torrent selectors
  const torrentSelectors = pattern.torrent ||
    SITE_PATTERNS.generic.torrent;
  for (const selector of torrentSelectors) {
    try {
      const elements = container.querySelectorAll(selector);

```

```

    for (const el of elements) {
      const href = el.getAttribute('href');
      if (href && isTorrentUrl(href)) {
        results.push({ element: el, type: 'torrent-file', url:
href, site: siteName });
      }
    }
  } catch (e) {
    // Invalid selector, skip
  }
}

return results;
}

```

---

## 11. Performance Optimization

### 11.1 Debounced Scanning

**Claim:** Debouncing prevents excessive DOM scans by delaying execution until a burst of mutations settles, typically 300-500ms after the last mutation. **Source:** CSS-Tricks - Debouncing and Throttling Explained **URL:** <https://css-tricks.com/debouncing-throttling-explained-examples/> **Date:** 2026-02-05 **Excerpt:** Debounce: Grouping a sudden burst of events into a single one **Confidence:** high

```

/**
 * Creates a debounced function that delays invoking callback
 * until after wait milliseconds
 * have elapsed since the last time the debounced function was
 * invoked.
 */
function debounce(callback, wait = 300, options = {}) {
  const { leading = false, trailing = true, maxWait } = options;
  let timer = null;
  let lastCall = 0;
  let lastInvoke = 0;
  let lastArgs;
  let lastThis;

  function invoke(time) {
    lastInvoke = time;
    const args = lastArgs;
    const ctx = lastThis;
    lastArgs = lastThis = null;
    callback.apply(ctx, args);
  }
}

```

```

function debounced(...args) {
  const now = Date.now();
  const isFirstCall = lastCall === 0;
  lastCall = now;
  lastArgs = args;
  lastThis = this;

  if (timer) clearTimeout(timer);
  if (leading && isFirstCall) invoke(now);

  const remainingWait = maxWait
    ? Math.min(wait, maxWait - (now - lastInvoke))
    : wait;

  timer = setTimeout(() => {
    timer = null;
    if (trailing && lastArgs) invoke(Date.now());
    lastCall = 0;
  }, remainingWait);
}

debounced.cancel = () => {
  if (timer) clearTimeout(timer);
  timer = null;
  lastCall = 0;
  lastArgs = lastThis = null;
};

debounced.flush = () => {
  if (timer) {
    clearTimeout(timer);
    invoke(Date.now());
    timer = null;
  }
};

return debounced;
}

```

## 11.2 Throttled Scanning

```

/**
 * Creates a throttled function that only invokes callback at most
 * once per period.
 * Useful for scroll events and other high-frequency triggers.
 */
function throttle(callback, limit = 200) {
  let inThrottle = false;
  let trailingCall = null;

  function throttled(...args) {

```

```

    if (!inThrottle) {
      callback.apply(this, args);
      inThrottle = true;
      setTimeout(() => {
        inThrottle = false;
        if (trailingCall) {
          const { ctx, args: trailingArgs } = trailingCall;
          trailingCall = null;
          callback.apply(ctx, trailingArgs);
        }
      }, limit);
    } else {
      // Store trailing call to ensure last event is processed
      trailingCall = { ctx: this, args };
    }
  }

  throttled.cancel = () => {
    inThrottle = false;
    trailingCall = null;
  };

  return throttled;
}

```

## 11.3 Preventing Layout Thrashing

**Claim:** Layout thrashing occurs when JavaScript alternates between reading layout properties and writing style changes, forcing the browser to recalculate layout repeatedly. **Source:** webperf.tips - Layout Thrashing and Forced Reflows **URL:** <https://webperf.tips/tip/layout-thrashing/> **Date:** 2022-12-11 **Excerpt:**

Reading positioning and styling information can be quite fast if performed on a fully cached, styled and positioned Render Tree

**Confidence:** high

```

/**
 * Batch DOM reads and writes to prevent layout thrashing.
 * Separates measurement (read) from mutation (write) phases.
 */
class DOMScheduler {
  constructor() {
    this.readQueue = [];
    this.writeQueue = [];
    this.scheduled = false;
  }

  read(fn) {
    this.readQueue.push(fn);
    this.schedule();
  }

```

```

    }

    write(fn) {
        this.writeQueue.push(fn);
        this.schedule();
    }

    schedule() {
        if (this.scheduled) return;
        this.scheduled = true;

        requestAnimationFrame(() => {
            // Execute all reads first
            for (const fn of this.readQueue) {
                try { fn(); } catch (e) { /* ignore */ }
            }
            this.readQueue = [];

            // Execute all writes after reads
            for (const fn of this.writeQueue) {
                try { fn(); } catch (e) { /* ignore */ }
            }
            this.writeQueue = [];

            this.scheduled = false;
        });
    }
}

// Usage for marking found elements
const scheduler = new DOMScheduler();

function markElementAsFound(element) {
    scheduler.write(() => {
        element.classList.add('torrent-link-detected');
        element.setAttribute('data-torrent-found', 'true');
    });
}

```

## 11.4 Performance Budget

```

/**
 * Performance monitor that ensures scanning doesn't exceed time
 * budgets.
 * Yields control back to the browser if scan takes too long.
 */
class PerformanceBudget {
    constructor(maxMs = 16) { // 16ms = one frame at 60fps
        this.maxMs = maxMs;
        this.startTime = 0;
    }
}

```

```

begin() {
  this.startTime = performance.now();
}

hasTimeRemaining() {
  return (performance.now() - this.startTime) < this.maxMs;
}

/**
 * Yield control if budget is exhausted.
 * Returns a promise that resolves when work can continue.
 */
async yieldIfNeeded() {
  if (!this.hasTimeRemaining()) {
    await new Promise(resolve =>
      requestAnimationFrame(resolve));
    this.begin();
  }
}

/**
 * Scans nodes respecting performance budget.
 */
async function scanWithBudget(nodes, scanner, budget = new
  PerformanceBudget(16)) {
  const results = [];
  budget.begin();

  for (const node of nodes) {
    results.push(...scanner.scanAnchorLinks(node));
    await budget.yieldIfNeeded();

    results.push(...scanner.scanClickHandlers(node));
    await budget.yieldIfNeeded();
  }

  return results;
}

```

---

## 12. Complete Implementation

### 12.1 Master Scanner Class

```

/**
 * Complete torrent/magnet link scanner for browser extension
 * content scripts.
 * Handles static DOM, dynamic content, iframes, and Shadow DOM.

```

```

*
* Usage:
*   const scanner = new TorrentDOMScanner();
*   scanner.start();
*   scanner.on('found', (links) => console.log('Found:', links));
*/
class TorrentDOMScanner extends EventTarget {
  constructor(options = {}) {
    super();

    this.options = {
      debounceMs: 500,
      scanBudgetMs: 16,
      enableMutationObserver: true,
      enableShadowDOMScan: true,
      enableTextNodeScan: true,
      enableInfiniteScroll: true,
      ...options
    };

    this.mutationObserver = null;
    this.iframeObserver = null;
    this.infiniteScrollObserver = null;
    this.linkScanner = new LinkScanner();
    this.foundLinks = new Map(); // URL -> metadata
                                (deduplication)
    this.isRunning = false;
    this.scheduler = new DOMScheduler();
  }

  /**
   * Start scanning the page.
   */
  start() {
    if (this.isRunning) return;
    this.isRunning = true;

    // Initial full scan
    this.performFullScan();

    // Set up MutationObserver for dynamic content
    if (this.options.enableMutationObserver) {
      this.setupMutationObserver();
    }

    // Set up infinite scroll detection
    if (this.options.enableInfiniteScroll) {
      this.setupInfiniteScrollDetection();
    }
  }

  /**

```

```

    * Stop scanning.
    */
stop() {
    this.isRunning = false;

    if (this.mutationObserver) {
        this.mutationObserver.disconnect();
        this.mutationObserver = null;
    }

    if (this.infiniteScrollObserver) {
        this.infiniteScrollObserver.disconnect();
        this.infiniteScrollObserver = null;
    }

    if (this.iframeObserver) {
        this.iframeObserver.disconnect();
        this.iframeObserver = null;
    }
}

/**
* Perform a complete scan of the current DOM.
*/
performFullScan() {
    const results = [];

    // 1. Scan with site-specific patterns
    results.push(...scanWithSitePatterns(document));

    // 2. Generic anchor link scan (fallback)
    results.push(...this.linkScanner.scanAnchorLinks(document));

    // 3. Click handler scan
    results.push(...this.linkScanner.scanClickHandlers(document));

    // 4. Text node scan for plain-text magnet links
    if (this.options.enableTextNodeScan) {
        const textResults = scanTextNodesForMagnets(document.body);
        for (const tr of textResults) {
            for (const magnet of tr.magnets) {
                results.push({
                    element: tr.element,
                    type: 'magnet',
                    url: magnet,
                    source: 'text-node'
                });
            }
        }
    }

    // 5. Shadow DOM scan

```



```

    if (this.options.enableShadowDOMScan) {
      results.push(...scanThroughShadowDOM(document));
    }

    this.processResults(results);
  }

  /**
   * Set up MutationObserver for incremental scanning.
   */
  setupMutationObserver() {
    const debouncedIncrementalScan = debounce((addedNodes) => {
      const results = incrementalScan(addedNodes);
      this.processResults(results);
    }, this.options.debounceMs);

    this.mutationObserver =
      setupDynamicContentObserver((addedNodes) => {
        debouncedIncrementalScan(addedNodes);
      }, { debounceMs: this.options.debounceMs });
  }

  /**
   * Set up infinite scroll detection.
   */
  setupInfiniteScrollDetection() {
    this.infiniteScrollObserver = setupInfiniteScrollDetector(()
      => {
        // Force a full scan when approaching bottom (new content
        // likely loaded)
        this.performFullScan();
      });
  }

  /**
   * Process scan results, deduplicate, and emit events.
   */
  processResults(results) {
    const newLinks = [];

    for (const result of results) {
      const url = result.url || result.element?.getAttribute?.(
        'href');
      if (!url) continue;

      // Deduplicate
      if (this.foundLinks.has(url)) continue;

      const parsed = parseMagnetLink(url);
      const metadata = {
        url,
        type: result.type,

```

```

        source: result.source || 'dom-scan',
        site: result.site,
        element: result.element,
        parsed: parsed,
        timestamp: Date.now()
    };

    this.foundLinks.set(url, metadata);
    newLinks.push(metadata);
}

if (newLinks.length > 0) {
    this.dispatchEvent(new CustomEvent('found', { detail:
        newLinks }));
}
}

/**
 * Get all found links so far.
 */
getAllLinks() {
    return Array.from(this.foundLinks.values());
}

/**
 * Clear all found links (useful for page navigation within
    SPA).
 */
clear() {
    this.foundLinks.clear();
}
}

```

## 12.2 Content Script Entry Point

```

/**
 * content.js - Content script entry point for browser extension.
 * This runs in the ISOLATED world with access to chrome.* APIs.
 */

(function() {
    'use strict';

    // Prevent double-injection
    if (window.__TORRENT_SCANNER_INITIALIZED) return;
    window.__TORRENT_SCANNER_INITIALIZED = true;

    let scanner = null;

    /**
     * Initialize the scanner when DOM is ready.
     */

```

```

*/
function init() {
  scanner = new TorrentDOMScanner({
    debounceMs: 500,
    enableMutationObserver: true,
    enableShadowDOMScan: true,
    enableTextNodeScan: true,
    enableInfiniteScroll: true
  });

  // Handle found links
  scanner.addEventListener('found', (event) => {
    const links = event.detail;

    // Send to background script
    chrome.runtime.sendMessage({
      action: 'torrentLinksFound',
      data: links.map(l => ({
        url: l.url,
        type: l.type,
        source: l.source,
        hash: l.parsed?.hash,
        name: l.parsed?.name
      })))
    }).catch(err => {
      // Extension may not be listening yet
    });

    // Optional: highlight found links on page
    highlightLinks(links);
  });

  scanner.start();

  // Listen for messages from popup/background
  chrome.runtime.onMessage.addListener((message, sender,
    sendResponse) => {
    if (message.action === 'getFoundLinks') {
      sendResponse({ links: scanner.getAllLinks() });
      return true;
    }
    if (message.action === 'rescan') {
      scanner.clear();
      scanner.performFullScan();
      sendResponse({ status: 'rescanned' });
      return true;
    }
  });
}

/**

```

```

    * Highlight found torrent links on the page (optional visual
      feedback).
  */
  function highlightLinks(links) {
    for (const link of links) {
      if (link.element && link.element.style) {
        link.element.style.outline = '2px solid #4CAF50';
        link.element.style.outlineOffset = '2px';
      }
    }
  }

  // Initialize when DOM is ready
  if (document.readyState === 'loading') {
    document.addEventListener('DOMContentLoaded', init);
  } else {
    init();
  }
}());

```

## 12.3 Manifest V3 Configuration

```

{
  "manifest_version": 3,
  "name": "Torrent Link Scanner",
  "version": "1.0.0",
  "description": "Detects torrent and magnet links on web pages",
  "permissions": [
    "activeTab",
    "scripting"
  ],
  "host_permissions": [
    "<all_urls>"
  ],
  "content_scripts": [
    {
      "matches": ["<all_urls>"],
      "js": ["content.js"],
      "run_at": "document_idle",
      "all_frames": true,
      "match_about_blank": true
    }
  ],
  "web_accessible_resources": [
    {
      "resources": ["inject.js"],
      "matches": ["<all_urls>"]
    }
  ],
  "background": {

```

```

    "service_worker": "background.js"
  },
  "action": {
    "default_popup": "popup.html",
    "default_icon": {
      "16": "icon16.png",
      "48": "icon48.png",
      "128": "icon128.png"
    }
  }
}
}

```

---

## 13. Edge Cases and Error Handling

### 13.1 Common Edge Cases

```

/**
 * Edge case handlers for robust scanning.
 */
const EDGE_CASE_HANDLERS = {
  /**
   * Handle encoded magnet URLs in href.
   * Some sites URL-encode magnet links.
   */
  decodeEncodedMagnet(element) {
    const href = element.getAttribute('href');
    if (!href) return null;

    // Check if the href is URL-encoded
    if (href.includes('%3A') || href.includes('%3F')) {
      try {
        const decoded = decodeURIComponent(href);
        if (decoded.startsWith('magnet:')) {
          return decoded;
        }
      } catch (e) {
        // Invalid encoding, ignore
      }
    }
    return null;
  },

  /**
   * Handle magnet links in javascript: URLs.
   * e.g., href="javascript:window.location='magnet:?xt=...'"
   */
  extractFromJavaScriptUrl(element) {
    const href = element.getAttribute('href') || '';

```

```

    const jsMatch = href.match(/javascript:.*?(magnet:\?
      xt=urn:btih:[a-f0-9]{32,40}[^"\s]*)/i);
    return jsMatch ? jsMatch[1] : null;
  },

  /**
   * Handle Base32-encoded BTIH hashes (32 chars).
   * Normal hex hashes are 40 chars.
   */
  normalizeHash(hash) {
    if (hash.length === 40) {
      return hash.toLowerCase(); // Hex hash
    }
    if (hash.length === 32) {
      return hash.toUpperCase(); // Base32 hash
    }
    return hash;
  },

  /**
   * Handle links that use data-* attributes instead of href.
   */
  extractFromDataAttributes(element) {
    for (const attr of element.attributes) {
      if (attr.name.startsWith('data-') &&
        attr.value.startsWith('magnet:')) {
        return attr.value;
      }
      // Some sites store the hash only
      if (attr.name === 'data-hash' && attr.value.length === 40) {
        return `magnet:?xt=urn:btih:${attr.value}`;
      }
    }
    return null;
  },

  /**
   * Prevent duplicate detection across multiple scan passes.
   */
  deduplicateKey(url) {
    // Normalize URL for deduplication
    try {
      const parsed = parseMagnetLink(url);
      if (parsed) {
        return `magnet:${parsed.hash}`;
      }
    } catch (e) {}
    return url;
  }
};

```

## 13.2 Error Handling

```
/**
 * Wraps any function with error handling that won't break the
 * scanner.
 */
function safeExecute(fn, fallback = null, context = 'unknown') {
  try {
    return fn();
  } catch (error) {
    console.warn(`[TorrentScanner] Error in ${context}:`, error);
    return fallback;
  }
}

/**
 * Safe querySelectorAll that won't throw on invalid selectors.
 */
function safeQuerySelectorAll(container, selector) {
  return safeExecute(
    () => container.querySelectorAll(selector),
    [],
    `querySelectorAll("${selector}")`
  );
}

/**
 * Safe TreeWalker creation.
 */
function safeCreateTreeWalker(root, whatToShow, filter) {
  return safeExecute(
    () => document.createTreeWalker(root, whatToShow, filter,
      false),
    null,
    'createTreeWalker'
  );
}
```

## 13.3 Memory Leak Prevention

```
/**
 * Cleanup utilities to prevent memory leaks in long-running
 * content scripts.
 */
class ScannerCleanup {
  constructor() {
    this.observers = [];
    this.timers = [];
    this.listeners = [];
  }
}
```

```

trackObserver(observer) {
  this.observers.push(observer);
  return observer;
}

trackTimer(timer) {
  this.timers.push(timer);
  return timer;
}

trackListener(target, type, handler) {
  target.addEventListener(type, handler);
  this.listeners.push({ target, type, handler });
}

/**
 * Call this when the scanner is being destroyed or page is
 * navigating.
 */
cleanupAll() {
  for (const observer of this.observers) {
    try { observer.disconnect(); } catch (e) {}
  }
  this.observers = [];

  for (const timer of this.timers) {
    try { clearTimeout(timer); clearInterval(timer); }
    catch (e) {}
  }
  this.timers = [];

  for (const { target, type, handler } of this.listeners) {
    try { target.removeEventListener(type, handler); }
    catch (e) {}
  }
  this.listeners = [];
}
}

```

---

## 14. Test Cases

### 14.1 HTML Test Fixtures

*<!-- Test Case 1: Basic magnet link in <a> tag -->*

```

<a href="magnet:
  xt=urn:btih:211361b71d2e589ca44b99e0b9ce7d838d58e48a&dn=Ubuntu&tr=udp://
  tracker.example.com">Download Ubuntu</a>

```

*<!-- Test Case 2: Magnet link as plain text -->*



```
<code>magnet:?
xt=urn:btih:42f7d12bdf685907ddc7eae532d3e4214e8f12d5&dn=Example</
code>
```

```
<!-- Test Case 3: .torrent file link -->
```

`<a href="https://example.com/file.torrent">Download .torrent</a>`

```
<!-- Test Case 4: Data attribute magnet -->
```

```
<button data-magnet="magnet:?
xt=urn:btih:37a8e3b4564996c13b408ebb695431dbf9f0e1c8">Copy
Magnet</button>
```

```
<!-- Test Case 5: JavaScript URL with magnet -->
```

```
<a href="javascript:window.location='magnet:xt=urn:btih:abc123...'">Open</a>
```

```
<!-- Test Case 6: URL-encoded magnet -->
```

<a href="magnet%3A%3Fxt%3Durn%3Aabtih%3A211361b71d2e589ca44b99e0b9ce7d838d58e48a">E

```
<!-- Test Case 7: Nested in Shadow DOM -->
```

```
<custom-element>
  <!-- Shadow DOM contains: <a href="magnet:?
    xt=urn:btih:...">Link</a> -->
```

&lt;/custom-element&gt;

```
<!-- Test Case 8: Invalid/truncated magnet (should NOT match) -->
```

**<span>magnet:?xt=urn:btih:T00\_SHORT</span>**

```
<!-- Test Case 9: Inside <pre> block -->
```

```


```

magnet:?xt=urn:btih:211361b71d2e589ca44b99e0b9ce7d838d58e48a&dn=Linux  
Some other text here

&lt;/pre&gt;

```
<!-- Test Case 10: Multiple magnets in one element -->
```

<div>

```
magnet:?xt=urn:btih:111111111111111111111111111111111111&dn=File1
```

and

```
magnet:?xt=urn:btih:222222222222222222222222222222222222&dn=File2
```

&lt;/div&gt;

## 14.2 JavaScript Test Suite

/\*\*

\* *Test suite for the torrent scanner.*

\* Can be run in browser console or with a test framework.

\* /

```
function runTests() {
  let passed = 0;
  let failed = 0;
```

```
function assert(condition, name) {
  if (condition) { passed++; console.log(`PASS: ${name}`); }
  else { failed++; console.error(`FAIL: ${name}`); }
}

// Test parseMagnetLink
const validMagnet = 'magnet:?
xt=urn:btih:211361b71d2e589ca44b99e0b9ce7d838d58e48a&dn=Ubuntu&tr=udp://
tracker.example.com';
const parsed = parseMagnetLink(validMagnet);
assert(parsed !== null, 'Valid magnet link parses correctly');
assert(parsed.hash ===
'211361b71d2e589ca44b99e0b9ce7d838d58e48a', 'Hash
extracted correctly');
assert(parsed.hashType === 'hex', 'Hash type detected as hex');
assert(parsed.name === 'Ubuntu', 'Display name extracted');
assert(parsed.trackers.length === 1, 'Tracker extracted');

// Test base32 hash (32 chars)
const base32Magnet = 'magnet:?
xt=urn:btih:FRVWQWGWUWUWIPKCGGTR3NFRZETEWBUF&dn=test';
const base32Parsed = parseMagnetLink(base32Magnet);
assert(base32Parsed !== null, 'Base32 hash magnet parses
correctly');
assert(base32Parsed.hashType === 'base32', 'Hash type detected
as base32');

// Test invalid magnets
assert(parseMagnetLink('http://example.com') === null, 'Non-
magnet URL rejected');
assert(parseMagnetLink('magnet:?xt=urn:btih:SHORT') === null,
'Short hash rejected');
assert(parseMagnetLink('magnet:not-a-valid-link') === null,
'Malformed magnet rejected');

// Test regex matching
const multiText = 'Here is magnet:?
xt=urn:btih:111111111111111111111111111111111111111111111111111&dn=A
and also magnet:?
xt=urn:btih:222222222222222222222222222222222222222222222222222&dn=B';
const matches = multiText.match(MAGNET_PATTERNS.FULL);
assert(matches && matches.length === 2, 'Multiple magnets in
text detected');

// Test .torrent detection
assert(isTorrentUrl('http://example.com/file.torrent') ===
true, 'Direct .torrent detected');
assert(isTorrentUrl('http://example.com/file.torrent?hash=abc')
=== true, '.torrent with query params');
assert(isTorrentUrl('http://example.com/download.php?id=123')
=== false, 'Non-torrent URL rejected');

// Test deduplication
```

```

const dedupMap = new Map();
const key1 = EDGE_CASE_HANDLERS.deduplicateKey(validMagnet);
const key2 = EDGE_CASE_HANDLERS.deduplicateKey(validMagnet +
    '&tr=extra');
assert(key1 === key2, 'Same magnet with different trackers
    deduplicated');

console.log(`\nTests: ${passed} passed, ${failed} failed`);
return { passed, failed };
}

```

---

## 15. Citations and Sources

### Primary Sources

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| 2 | MDN - content_scripts manifest          | https://developer.mozilla.org/en-US/docs/Mozilla/Add-ons/WebExtensions/manifest.json/content_scripts | 2025-11-26 | S (MDN)      |
| 3 | MDN - Using shadow DOM                  | https://developer.mozilla.org/en-US/docs/Web/API/Web_components/Using_shadow_DOM                     | 2025-12-17 | S (MDN)      |
| 4 | Chrome for Developers - Content Scripts | https://developer.chrome.com/docs/extensions/develop/concepts/content-scripts                        | 2012-09-17 | S (Official) |
| 5 | Chrome - scripting API                  | https://developer.chrome.com/docs/extensions/reference/api/scripting                                 | 2026-01-08 | S (Official) |
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| #  | Source                             | URL                                                                                                                                                                                                                                 | Date       | Authority     |
|----|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------|
|    | MS Edge - Minimize page load time  | <a href="https://learn.microsoft.com/en-us/microsoft-edge/extensions/developer-guide/minimize-page-load-time-impact">https://learn.microsoft.com/en-us/microsoft-edge/extensions/developer-guide/minimize-page-load-time-impact</a> |            |               |
| 7  | W3C - DOM Traversal Spec           | <a href="https://www.w3.org/TR/DOM-Level-2-Traversal-Range/traversal.html">https://www.w3.org/TR/DOM-Level-2-Traversal-Range/traversal.html</a>                                                                                     | N/A        | S (W3C)       |
| 8  | CSS-Tricks - Debouncing/Throttling | <a href="https://css-tricks.com/debouncing-throttling-explained-examples/">https://css-tricks.com/debouncing-throttling-explained-examples/</a>                                                                                     | 2026-02-05 | A (Authority) |
| 9  | webperf.tips - Layout Thrashing    | <a href="https://webperf.tips/tip/layout-thrashing/">https://webperf.tips/tip/layout-thrashing/</a>                                                                                                                                 | 2022-12-11 | A (Authority) |
| 10 | BEP 0053 - Magnet URI extension    | <a href="https://www.bittorrent.org/beps/bep_0053.html">https://www.bittorrent.org/beps/bep_0053.html</a>                                                                                                                           | N/A        | S (Spec)      |
| 11 | Tixati Protocol v3.1               | <a href="https://tixati.com/specs/bittorrent/v3.1">https://tixati.com/specs/bittorrent/v3.1</a>                                                                                                                                     | 2025-10-07 | A (Spec)      |

## Implementation References

| #  | Source                     | URL                                                                                                                   | Date       | Authority  |
|----|----------------------------|-----------------------------------------------------------------------------------------------------------------------|------------|------------|
| 12 | magnet-link-regex npm      | <a href="https://github.com/tiaanduplessis/magnet-link-regex">https://github.com/tiaanduplessis/magnet-link-regex</a> | 2018-06-04 | S (GitHub) |
| 13 | parse-torrent (WebTorrent) | <a href="https://github.com/webtorrent/parse-torrent">https://github.com/webtorrent/parse-torrent</a>                 | 2014-02-24 | S (GitHub) |
| 14 | magnet-uri (WebTorrent)    | <a href="https://github.com/webtorrent/magnet-uri">https://github.com/webtorrent/magnet-uri</a>                       | 2013-10-28 | S (GitHub) |
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| 16 | Magnet2Deluge extension         | salesforce/kagekiri<br>https://github.com/jkctech/Magnet2Deluge                      | 2025-07-06 | S (GitHub)      |
| 17 | magnet-linker-browser-extension | https://github.com/trossr32/magnet-linker-browser-extension                          | 2019-07-03 | S (GitHub)      |
| 18 | Torrent Search extension        | https://webextension.org/listing/torrent-search.html                                 | 2024-08-04 | A (Product)     |
| 19 | Magnet Link Finder userscript   | https://gist.github.com/patik/8388768                                                | 2014-01-12 | S (GitHub Gist) |
| 20 | MutationObserver guide          | https://macarthur.me/posts/use-mutation-observer-to-handle-nodes-that-dont-exist-yet | 2023-03-13 | A (Blog)        |

## Summary

### Key Technical Decisions

1. **Use `document_idle` with DOM readiness checks** - The default timing is best for most cases, with fallback checks for `readyState`.
2. **Combine CSS selectors + TreeWalker + MutationObserver** - CSS selectors are fastest for known patterns, TreeWalker handles text nodes, MutationObserver catches dynamic content.
3. **Debounced incremental scanning (500ms)** - Batches rapid DOM changes to avoid excessive CPU usage while remaining responsive.

- 4. **Performance budget (16ms/frame)** - Yields control back to the browser to prevent UI jank on heavy pages.
- 5. **Deduplication by info hash** - Normalizes and deduplicates magnet links by their BTIH hash, not full URL.
- 6. **all\_frames: true + match\_about\_blank: true** - Ensures scanning inside iframes and dynamically created blank frames.
- 7. **Shadow DOM recursive traversal** - Manual recursion into shadowRoot properties since querySelectorAll doesn't pierce shadow boundaries.

Browser Compatibility

| Feature              | Chrome    | Firefox    | Safari      | Edge      |
|----------------------|-----------|------------|-------------|-----------|
| MutationObserver     | Yes (26+) | Yes (14+)  | Yes (7+)    | Yes (12+) |
| IntersectionObserver | Yes (51+) | Yes (55+)  | Yes (12.1+) | Yes (15+) |
| Shadow DOM v1        | Yes (53+) | Yes (63+)  | Yes (10.1+) | Yes (79+) |
| TreeWalker           | All       | All        | All         | All       |
| Manifest V3          | Yes       | Yes (101+) | Yes         | Yes       |

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